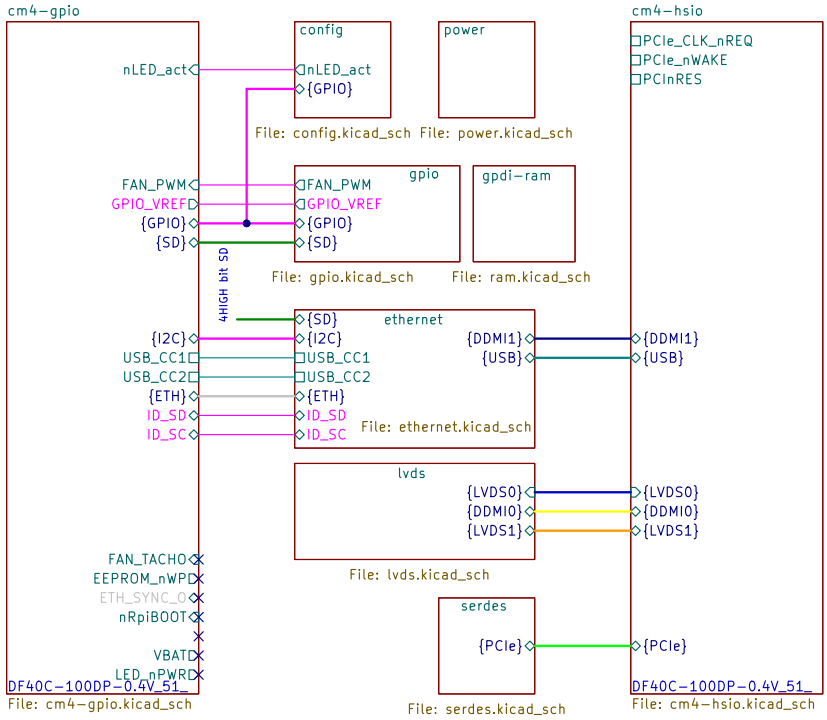
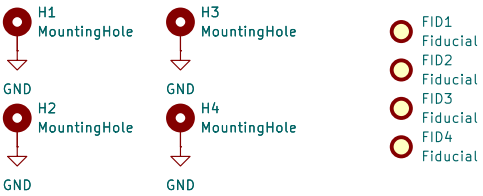
ULX5M-GS v006
Intergalaktik d.o.o.
License: CERN-OHL-S v2
Year: 2025



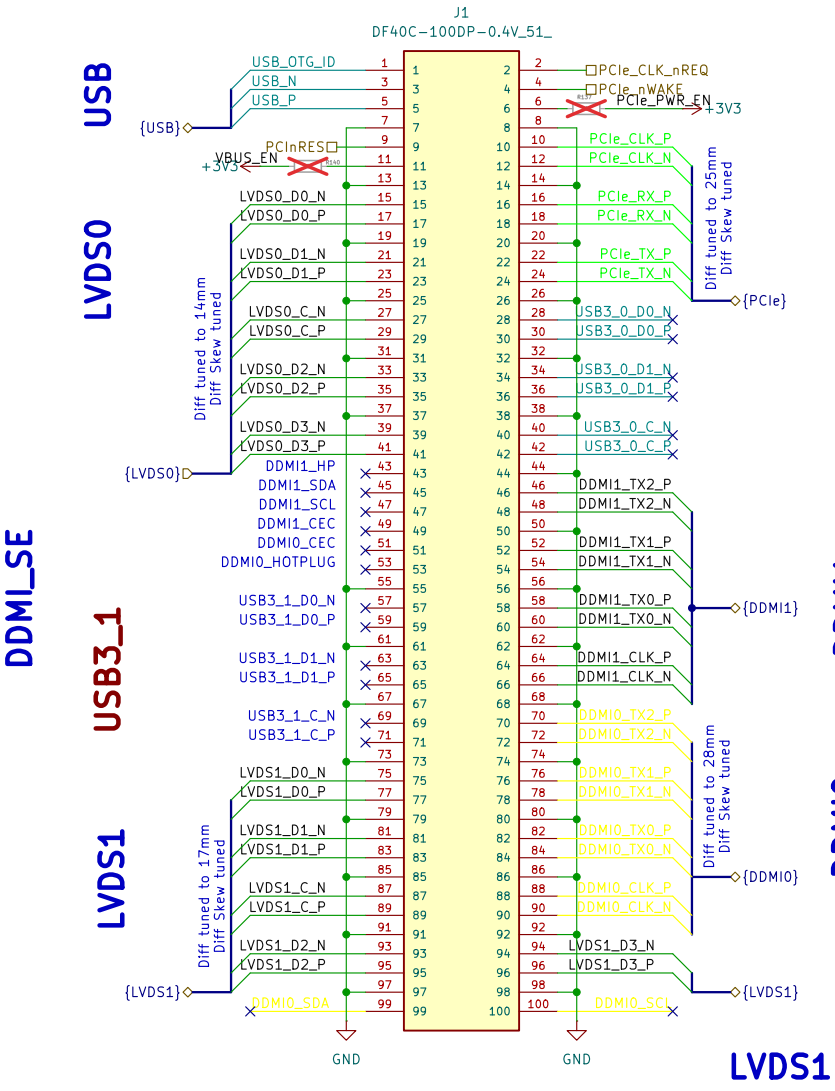


CERN-OHL-S v2
Intergalaktik d.o.o.

Sheet: /
File: ulx5m-gs.kicad_sch

Title: ULX5M-GS

Size: A5	Date: 2025-06-24	**Rev: v006**
KiCad E.D.A. 10.0.6		Id: 1/10



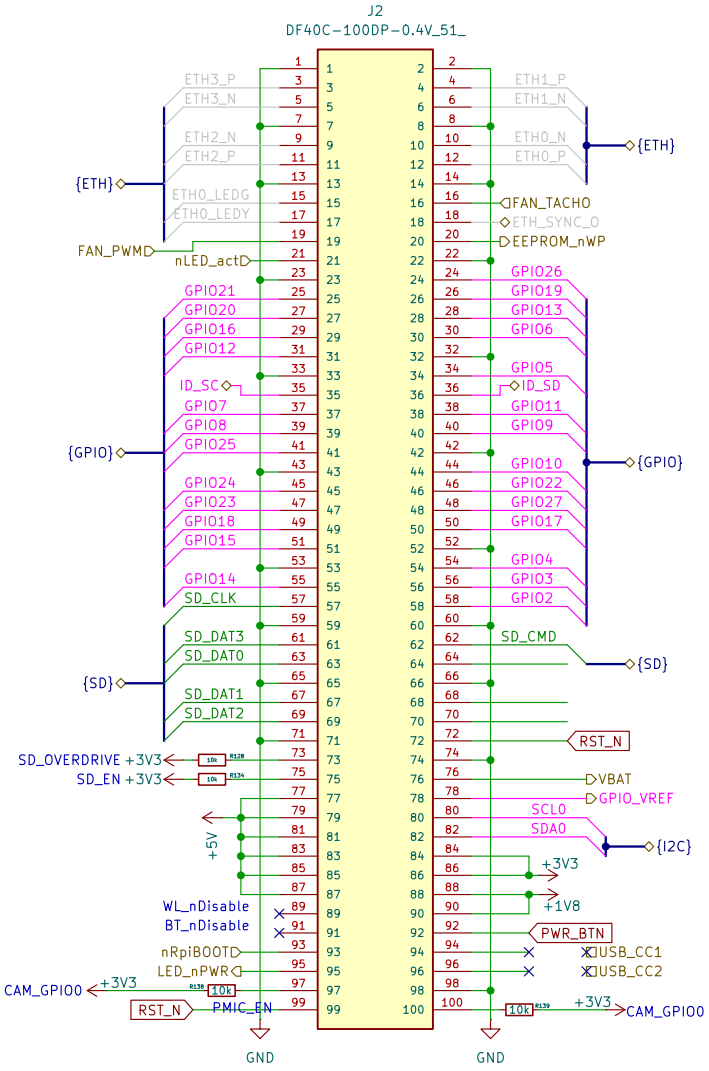
CERN-OHL-S v2

Intergalaktik d.o.o.

Sheet: /cm4-hsio/
File: cm4-hsio.kicad_sch

Title: ULX5M-GS

Size: A5	Date: 2025-06-24	**Rev: v006**
KiCad E.D.A. 10.0.6		Id: 2/10



CERN-OHL-S v2

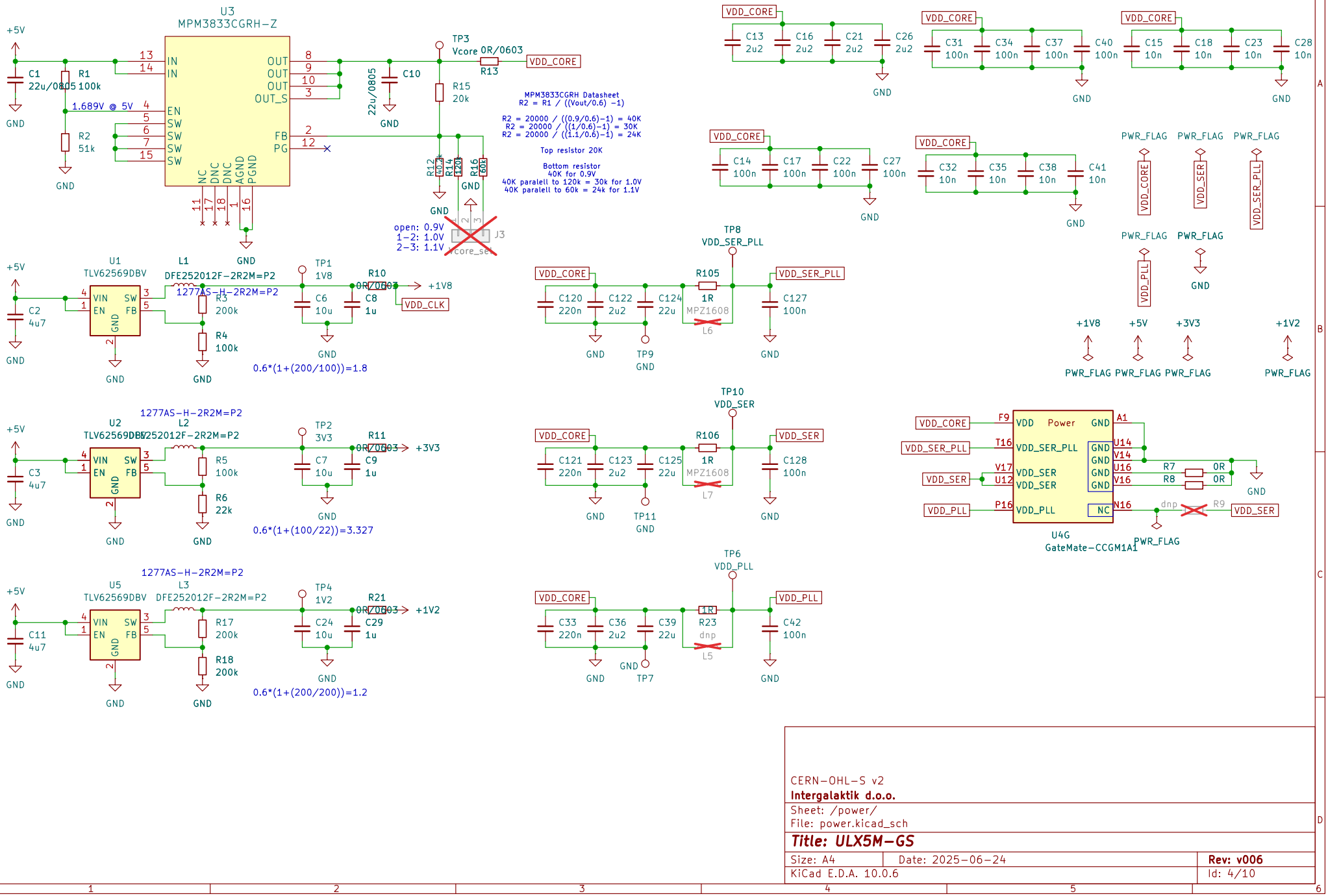
Intergalaktik d.o.o.

Sheet: /cm4-gpio/

File: cm4-gpio.kicad_sch

Title: ULX5M-GS

Size: A5	Date: 2025-06-24	**Rev: v006**
KiCad E.D.A. 10.0.6		Id: 3/10


U3
MPM3833CGRH-Z
C1
22u/0805
R1
100k
1.689V @ 5V
R2
51k
C10
22u/0805
TP3
Vcore 0R/0603
R13
R15
20k
VDD_CORE
MPM3833CGRH Datasheet
R2 = R1 / ((Vout/0.6) -1)
R2 = 20000 / ((0.9/0.6)-1) = 40K
R2 = 20000 / ((1/0.6)-1) = 30K
R2 = 20000 / ((1.1/0.6)-1) = 24K
Top resistor 20K
Bottom resistor
40K for 0.9V
40K paralell to 120k = 30k for 1.0V
40K paralell to 60k = 24k for 1.1V
R12 402k
R14 120k
R16 60k
J3
Vcore_sel
open: 0.9V
1-2: 1.0V
2-3: 1.1V
U1
TLV62569DBV
L1
DFE252012F-2R2M=P2
1277AS-H-2R2M=P2
R3 200k
R4 100k
TP1
1V8
R10
0R/0603
+1V8
VDD_CLK
C2 4u7
C6 10u
C8 1u
0.6*(1+(200/100))=1.8
U2
TLV62569DBV
L2
1277AS-H-2R2M=P2
R5 100k
R6 22k
TP2
3V3
R11
0R/0603
+3V3
C3 4u7
C7 10u
C9 1u
0.6*(1+(100/22))=3.327
U5
TLV62569DBV
L3
DFE252012F-2R2M=P2
1277AS-H-2R2M=P2
R17 200k
R18 200k
TP4
1V2
R21
0R/0603
+1V2
C11 4u7
C24 10u
C29 1u
0.6*(1+(200/200))=1.2
C13 2u2
C16 2u2
C21 2u2
C26 2u2
C31 100n
C34 100n
C37 100n
C40 100n
C15 10n
C18 10n
C23 10n
C28 10n
C14 100n
C17 100n
C22 100n
C27 100n
C32 10n
C35 10n
C38 10n
C41 10n
TP8
VDD_SER_PLL
R105
1R
MPZ1608
L6
C120 220n
C122 2u2
C124 22u
C127 100n
TP9
TP10
VDD_SER
R106
1R
MPZ1608
L7
C121 220n
C123 2u2
C125 22u
C128 100n
TP11
TP6
VDD_PLL
R23
1R
dnp
L5
C33 220n
C36 2u2
C39 22u
C42 100n
TP7
PWR_FLAG
VDD_CORE
VDD_SER
VDD_SER_PLL
VDD_PLL
+1V8
+5V
+3V3
+1V2
GND
U4G
GateMate-CCGM1A1
Power
VDD
VDD_SER_PLL
VDD_SER
VDD_PLL
R7 0R
R8 0R
R9 dnp
F9
T16
V17
U12
P16
A1
U14
V14
U16
V16
N16
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /power/
File: power.kicad_sch
Title: ULX5M-GS
Size: A4
Date: 2025-06-24
Rev: v006
KiCad E.D.A. 10.0.6
Id: 4/10



Use MD0/1 PU/PD for CPHA/CPOL selection

SW1 29-CVS-02B

U4B GateMate-CCGM1A1

U6 74LVC1G04GM,115

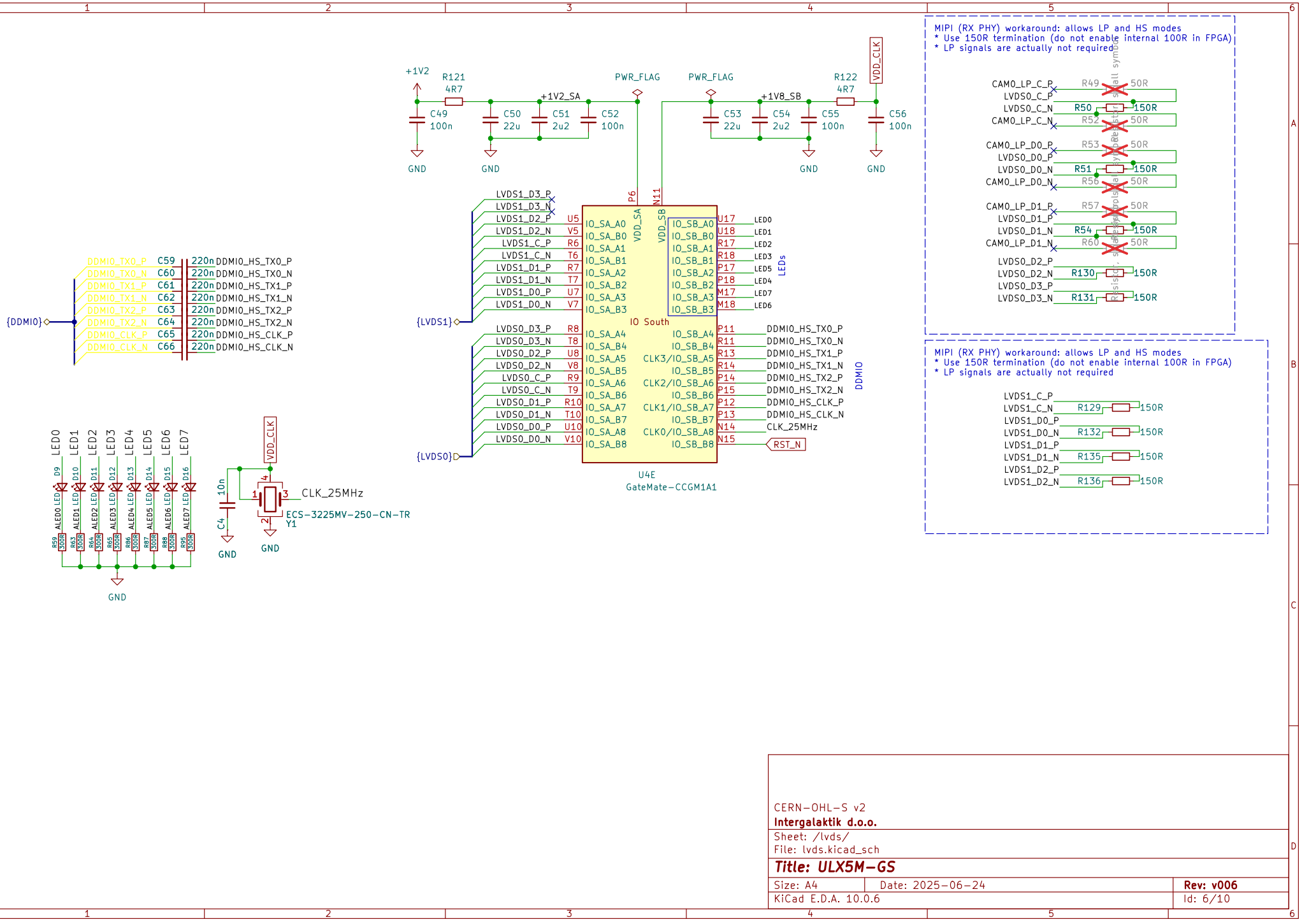
U7 74LVC2G00HK3-7

U8 74LVC2G00HK3-7

U9 AT25QL321-MBUE-T

J4 JTAG

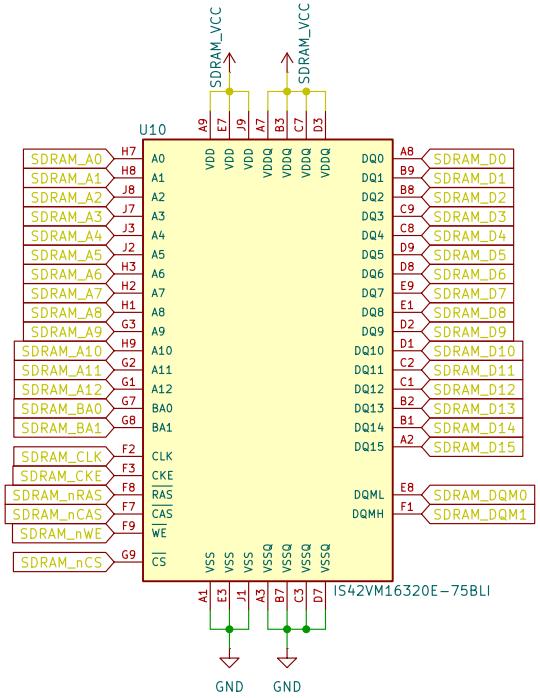
CERN-OHL-S v2	
Intergalaktik d.o.o.	
Sheet: /config/	
File: config.kicad_sch	
Title: ULX5M-GS	
Size: A4	Date: 2025-06-24
KiCad E.D.A. 10.0.6	Id: 5/10
	Rev: v006

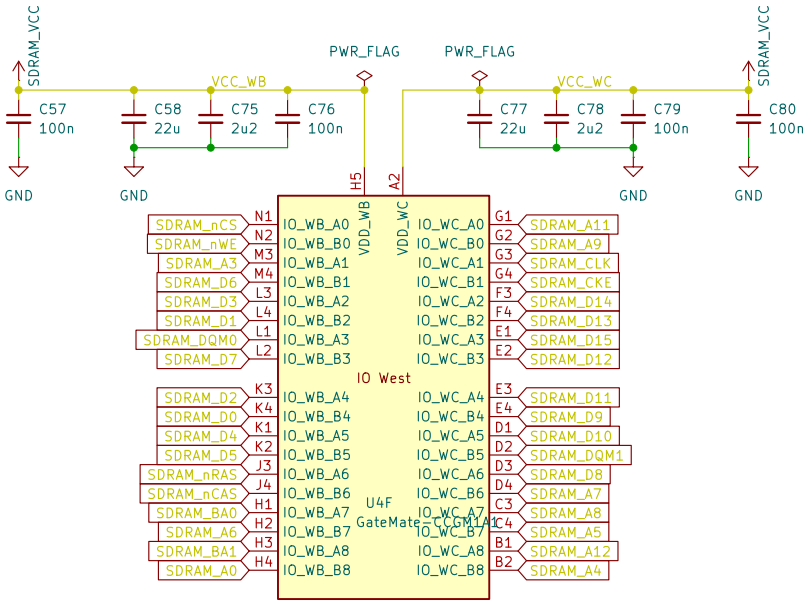


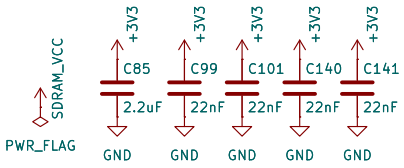
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /lvds/
File: lvds.kicad_sch

Title: ULX5M-GS

Size: A4	Date: 2025-06-24	**Rev: v006**
KiCad E.D.A. 10.0.6		Id: 6/10







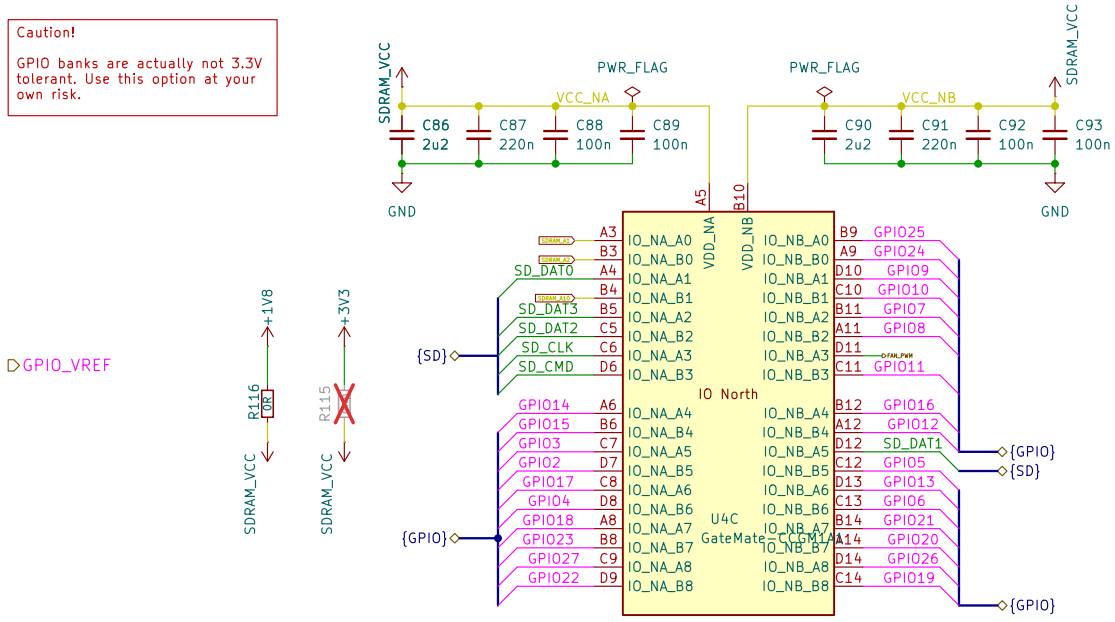
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /gpdi-ram/
File: ram.kicad_sch

Title: ULX5M-GS

Size: A4	Date: 2025-06-24	**Rev: v006**
KiCad E.D.A. 10.0.6		Id: 7/10

Caution!

GPIO banks are actually not 3.3V tolerant. Use this option at your own risk.



CERN-OHL-S v2

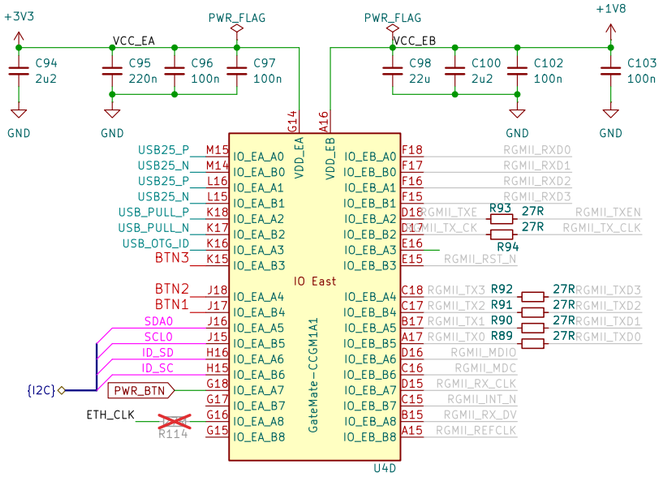
Intergalaktik d.o.o.

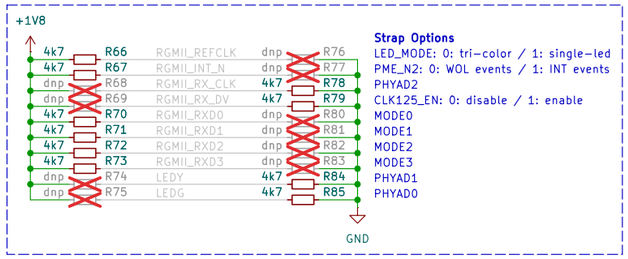
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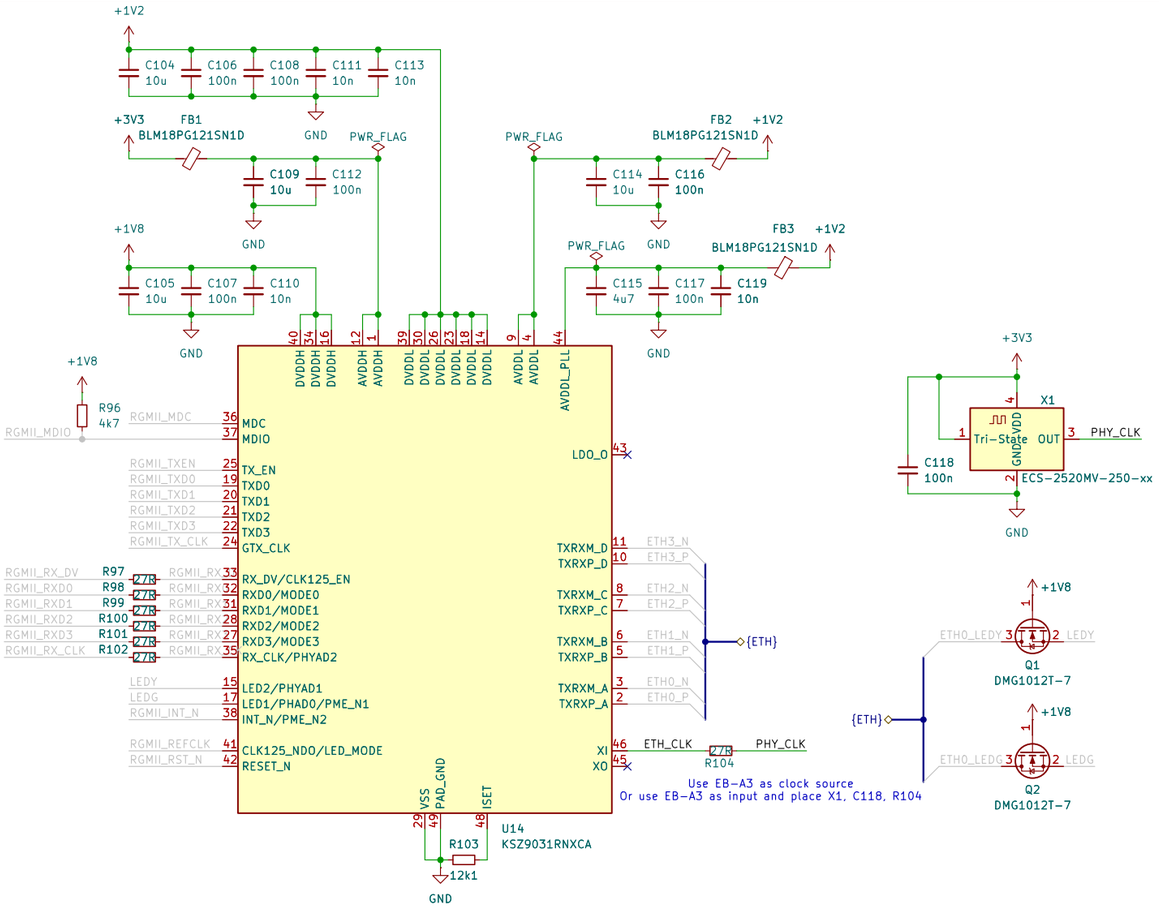
File: gpio.kicad_sch

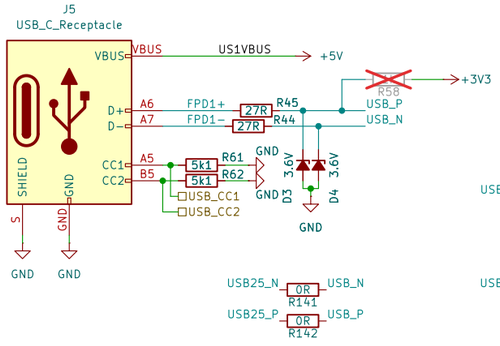
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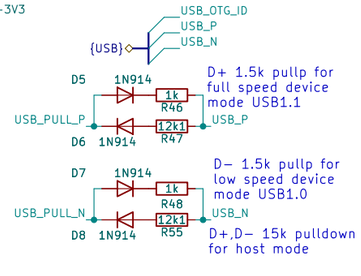
Size: A4	Date: 2025-06-24	**Rev: v006**
KiCad E.D.A. 10.0.6		Id: 8/10

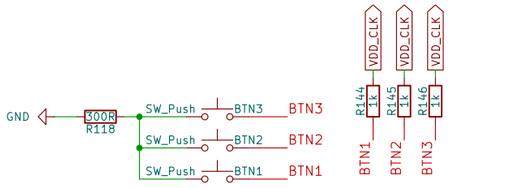








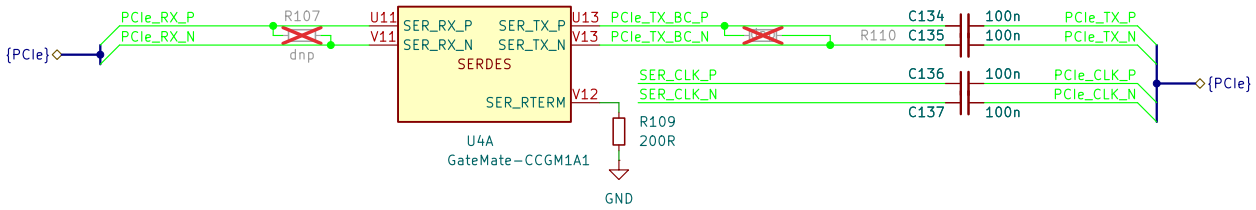


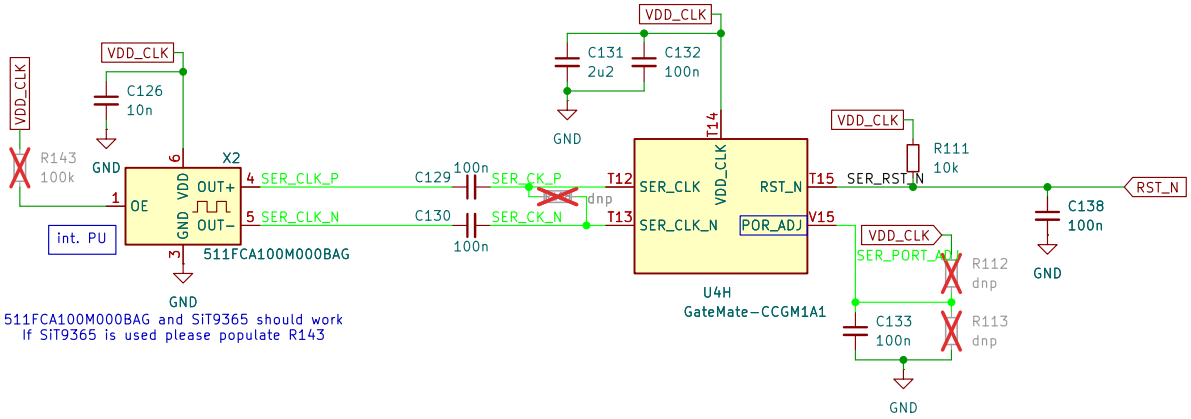


CERN-OHL-S v2

Intergalaktik d.o.o.		
Sheet: /ethernet/ File: ethernet.kicad_sch		
Title: ULX5M-GS		
Size: A3	Date: 2025-06-24	**Rev: v006**
KiCad E.D.A. 10.0.6		Id: 9/10

Note!: Take in mind that SerDes is not working when VDD_CORE = 0.9V (Low power mode), because VDDSER
and VDDSER_PLL have got minimum working voltage: 0.95V !





CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /serdes/
File: serdes.kicad_sch

Title: ULX5M-GS

Size: A4	Date: 2025-06-24	**Rev: v006**
KiCad E.D.A. 10.0.6		Id: 10/10